A Multiplicative-Orbit Continuation for the Alaoglu–Erdős Conjecture

Exact $\times 2, \times 3$ Carry Rectangles, Universal Smooth Residues, Digit-Lift Defects, and the Remaining Quantitative Rigidity Target

OpenAI GPT-5.6 Pro

Independent continuation of the supplied unified working report

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Abstract

The Alaoglu–Erdős conjecture asserts that a real number x satisfying $2^x, 3^x \in \mathbb{Z}$ must be an integer. The supplied working report already develops a large collection of exact reductions, determinant methods, p-adic audits, Sturmian codings, interpolation schemes, and formalization plans. This continuation deliberately avoids repeating those arguments. Its main new object is the *two-dimensional multiplicative carry field* associated with the semigroup orbit $\{2^r 3^s x\}$.

For every $R \times S$ rectangle, the horizontal binary carries and vertical ternary carries satisfy an exact plaquette equation. We prove a complete classification: there are precisely $2^R 3^S$ compatible carry rectangles, one for each cylinder of length $(2^R 3^S)^{-1}$, although the unconstrained edge alphabet contains $2^{R(S+1)} 3^{(R+1)S}$ labelings. Thus compatibility retains exactly the fraction 6^{-RS} of the naive labelings. Furstenberg's $\times 2, \times 3$ density theorem then implies that every compatible finite rectangle occurs, arbitrarily far out, for *every* irrational exponent. This is both a useful structural theorem and a sharp negative result: no proof based only on finitely forbidden carry patterns can settle the conjecture.

Applied to a hypothetical counterexample, the same theorem yields complete residue classes modulo every 3-smooth modulus for the two synchronized reduced-denominator exponents. The extreme all-zero rectangles give arbitrarily long, exactly scaled blocks and simultaneous leading zero gaps in the binary expansion of $(2^x)^q$ and the ternary expansion of $(3^x)^q$. We compare the first such block with Baker-type lower bounds and with the best published effective $\times 2, \times 3$ density bounds. The resulting corridor is enormous: polynomial from below and roughly triple-exponential from above. A precise sufficient closing lemma is isolated: simultaneous integrality would have to force subpolynomial smooth hitting times along an unbounded sequence of 3-smooth moduli.

A second new package introduces the binary-to-ternary digit lift $L(\sum \varepsilon_j 2^j) = \sum \varepsilon_j 3^j$ and its ternary-to-binary companion. Strict power inequalities, quantitative defects, composition laws, valuation transfer, and tail constraints are proved. These produce a new family of integral digit defects for every power of a hypothetical output pair, but no contradiction is claimed. The report concludes with reproducible computations, a Lean module graph, and a prioritized list of exact next targets. The conjecture remains open in this report.

Contents

1 Executive result and claim discipline

1.1 Outcome

This report does *not* claim a proof of the Alaoglu–Erdős conjecture. It establishes two new exact theorem packages, proves several consequences for any hypothetical counterexample, rules out a broad class of finite-symbolic strategies, and isolates a quantitative lemma which would close the problem if proved.

The central conclusions are as follows.

1. **Exact finite carry classification** (NEW–PROVED): compatible $R \times S$ binary–ternary carry rectangles are in bijection with the $2^R 3^S$ half-open cylinders of denominator $2^R 3^S$.
2. **Universal finite language** (CLASSICAL + NEW–PROVED): Furstenberg density implies that every compatible finite carry rectangle occurs for every irrational exponent, and occurs beyond every prescribed pair of semigroup coordinates.
3. **Complete smooth floor residues** (NEW–PROVED from Furstenberg): for every irrational α and every $Q = 2^R 3^S$,
$$\left\{ \lfloor 2^r 3^s \alpha \rfloor \bmod Q : r \geq R,\ s \geq S \right\} = \mathbb{Z}/Q\mathbb{Z}.$$
4. **Synchronized denominator universality** (NEW–PROVED, conditional on a counterexample): the reduced denominator exponents of the rational numbers $2^{\{2^r 3^s x\}}$ and $3^{\{2^r 3^s x\}}$ simultaneously attain every residue modulo every 3-smooth modulus.
5. **Finite-pattern no-go theorem** (NO–GO): a local rule that uses only bounded carry windows cannot distinguish a hypothetical counterexample from an arbitrary irrational exponent.
6. **Quantitative corridor** (CLASSICAL + NEW–PROVED): for the first all-zero block of modulus Q, Baker theory gives a polynomial lower bound, while currently published effective Furstenberg bounds give only a triple-exponential-scale upper bound.
7. **Digit-lift inequalities** (NEW–PROVED): binary digits evaluated at 3 lie strictly below the corresponding real power, while ternary digits evaluated at 2 lie strictly above the inverse power. This gives positive integral defect sequences and exact tail inequalities for every hypothetical output pair.

Status: precise research status. The carry and digit-lift theorems are proved in full below. Furstenberg density and linear-forms-in-logarithms estimates are external classical inputs. The proposed subpolynomial hitting-time statement is explicitly labeled as an unproved target. No numerical experiment is used as evidence for the conjecture itself.

1.2 Evidence labels

Label	Meaning
NEW–PROVED	A theorem proved in this report by a complete mathematical argument.
CLASSICAL	An external theorem used with a bibliographic citation.
INHERITED	A fact taken from the supplied unified report and used only as baseline.
COMPUTED	A finite diagnostic calculation; not a proof of an infinite claim.
TARGET	A precisely stated missing lemma or research program.
UNVERIFIED	A current public claim for which no adequate corroborating source was found.
NO–GO	A theorem showing that a proposed class of strategies cannot suffice.

1.3 Source identity and non-duplication boundary

The attached source was decompressed from `alaoglu-erdos-unified-report.tex.gz`. The decompressed source contains 20,483 lines and approximately 0.975 MB of text. It already treats, among many other themes, exact conditional solution monoids, primitive output pairs, the four-exponentials wall, rational-function rigidity, perfect powers, valuation lattices, generalized Vandermonde determinants, p-adic determinant ceilings, one-dimensional synchronized Sturmian words, compact S-integral orbits, Mahler and Padé diagnostics, moment methods, transfer spectra, additive carry languages, and a large Lean inventory [1].

The present report does not restate those developments. In particular:

- the source's Sturmian coding concerns the additive orbit $k \mapsto \{k\alpha\}$, whereas the new coding concerns the genuinely higher-rank multiplicative orbit $(r, s) \mapsto \{2^r 3^s \alpha\}$;
- the source's "carry language" is an additive rewrite-path construction, whereas the new carries are literal floor carries satisfying a two-dimensional plaquette law;
- determinant, interpolation, and p-adic ceiling calculations are cited only when explaining why the new route is complementary.

2 Dated verification of the motivating external claims

The user's motivating comparison involved several very recent claims. Because none of them is a mathematical premise of the arguments below, they are recorded only to establish an accurate research context as of 22 August 2026.

Claim	Status used here	Verification note
Claude Fable 5 exists and was publicly deployed	corroborated	Anthropic announced Fable 5 on 9 June 2026 and recorded redeployment/availability on 1 July 2026 [8].

Claim	Status used here	Verification note
"Resolved the Jacobian conjecture"	qualified	A 2026 arXiv paper records a counterexample in dimension three and constructions in every dimension greater than two. Thus the universal conjecture is refuted, while the paper does not settle the separate two-dimensional case [9].
Hadamard matrices for all previously unresolved orders below 2000	corroborated at announcement level	The OEIS historical note records constructions for the remaining twelve multiples of four at most 2000 and lists the orders explicitly [10].
Elliptic curve of rank at least 30	UNVERIFIED	The public item located for this claim described it as community discussion "not yet corroborated"; no certificate or primary paper was found in the search used for this report [11].
Recent large-scale Lean-assisted mathematics	corroborated in another case	A reported Anthropic project on the Riemann hypothesis used many coordinated agents and was checked in Lean; this demonstrates a workflow, not a result about the present conjecture [12].
Fable breakthrough on Alaoglu–Erdős	UNVERIFIED	Searches of public news, arXiv, and indexed web sources through 22 August 2026 found no public theorem, preprint, Lean repository, or proof certificate matching that claim. It is therefore not used as evidence or as a hidden lemma.

Audit warning 2.1. Recent announcements can be mathematically important without supplying a reusable lemma. The rest of this report depends only on explicit statements that can be proved below or cited to stable mathematical literature.

3 Minimal inherited baseline

We use only the following small portion of the supplied report's conditional structure. Assume, for contradiction, that a nonintegral solution exists. Passing to the primitive least positive generator supplied there, write

$$\beta \in \mathbb{R} \setminus \mathbb{Z}, \qquad M = 2^{\beta} \in \mathbb{N}, \qquad N = 3^{\beta} \in \mathbb{N}. \tag{1}$$

The inherited results imply:

1. β is irrational, indeed transcendental;
2. neither M is a power of 2 nor N a power of 3;
3. if
$$M = 2^{\lambda} W, \quad W \text{ odd}, \qquad N = 3^{\mu} Z, \quad 3 \nmid Z, \tag{2}$$
then
$$\min(\lambda, \mu) = 0; \tag{3}$$
4. for every $q = 2^r 3^s$, both $M^q = 2^{q\beta}$ and $N^q = 3^{q\beta}$ are integers.

These are labeled INHERITED. Everything beginning with the carry cocycle in Section 4 is developed independently here.

For later use define, for $r, s \geq 0$,

$$q_{r,s} = 2^r 3^s, \qquad \nu_{r,s} = \lfloor q_{r,s}\beta \rfloor, \qquad t_{r,s} = \{q_{r,s}\beta\}. \tag{4}$$

Then

$$X_{r,s} = \frac{M^{q_{r,s}}}{2^{\nu_{r,s}}} = 2^{t_{r,s}} \in \mathbb{Q} \cap [1,2), \qquad Y_{r,s} = \frac{N^{q_{r,s}}}{3^{\nu_{r,s}}} = 3^{t_{r,s}} \in \mathbb{Q} \cap [1,3). \tag{5}$$

The source report already studies the compact rational arc represented by (5). Our new contribution is the exact two-dimensional carry geometry of the indices $\nu_{r,s}$.

4 The multiplicative floor-carry cocycle

4.1 One-step carries and the cocycle identity

For an integer $m \geq 2$ and $t \in [0,1)$, put

$$T_m(t) = \{mt\}, \qquad c_m(t) = \lfloor mt \rfloor \in \{0, 1, \dots, m-1\}. \tag{6}$$

Thus

$$mt = c_m(t) + T_m(t). \tag{7}$$

The elementary identity below is the source of every compatibility equation in this report.

Lemma 4.1 (Multiplicative carry cocycle)**.** *For positive integers m, n and $t \in [0,1)$,*

$$T_{mn}(t) = T_n(T_m(t)) = T_m(T_n(t)), \tag{8}$$

$$c_{mn}(t) = nc_m(t) + c_n(T_m(t)) \tag{9}$$

$$= mc_n(t) + c_m(T_n(t)). \tag{10}$$

Proof. Write $mt = c_m(t) + T_m(t)$ and multiply by n:

$$mnt = nc_m(t) + nT_m(t) = nc_m(t) + c_n(T_m(t)) + T_n(T_m(t)).$$

The integer and fractional parts give (9) and the first equality in (8). Interchanging m and n gives (10) and the second equality. □

For $(m, n) = (2, 3)$, equality of the two expressions for $c_6(t)$ is

$$3c_2(t) + c_3(T_2 t) = 2c_3(t) + c_2(T_3 t). \tag{11}$$

This is a discrete curl-free law. The point is not that one plaquette is difficult; it is that the whole finite carry surface can be classified exactly.

4.2 Rectangular potentials and edge labels

Fix integers $R, S \geq 0$ and set

$$Q = 2^R 3^S. \tag{12}$$

For $t \in [0,1)$ define the vertex potentials

$$P_{a,b}(t) = \left\lfloor 2^a 3^b t \right\rfloor, \qquad 0 \leq a \leq R, \quad 0 \leq b \leq S. \tag{13}$$

The horizontal and vertical edge labels are

$$h_{a,b}(t) = P_{a+1,b}(t) - 2P_{a,b}(t) = c_2(T_{2^a 3^b} t) \in \{0,1\}, \tag{14}$$
$$v_{a,b}(t) = P_{a,b+1}(t) - 3P_{a,b}(t) = c_3(T_{2^a 3^b} t) \in \{0,1,2\}. \tag{15}$$

Here $h_{a,b}$ is defined for $0 \leq a < R$, $0 \leq b \leq S$, and $v_{a,b}$ for $0 \leq a \leq R$, $0 \leq b < S$.

Proposition 4.2 (Plaquette compatibility)**.** *Every elementary square satisfies*

$$3h_{a,b} + v_{a+1,b} = 2v_{a,b} + h_{a,b+1} \qquad (0 \leq a < R,\ 0 \leq b < S). \tag{16}$$

Proof. Compute the upper-right potential along the two directed paths:

$$P_{a+1,b+1} = 3P_{a+1,b} + v_{a+1,b} = 6P_{a,b} + 3h_{a,b} + v_{a+1,b},$$
$$P_{a+1,b+1} = 2P_{a,b+1} + h_{a,b+1} = 6P_{a,b} + 2v_{a,b} + h_{a,b+1}.$$

Equating them gives (16). □

Definition 4.3 (Compatible carry rectangle)**.** An $R \times S$ carry rectangle is a pair of arrays

$$h_{a,b} \in \{0,1\}, \qquad v_{a,b} \in \{0,1,2\}$$

on the edge sets above. It is *compatible* if every plaquette satisfies (16).

4.3 The exact cylinder–rectangle bijection

Partition $[0,1)$ into the Q half-open cylinders

$$I_k = \left[\frac{k}{Q}, \frac{k+1}{Q}\right), \qquad 0 \leq k < Q. \tag{17}$$

The next theorem is the finite combinatorial core of the report.

Theorem 4.4 (Exact carry-rectangle classification)**.** *Let $R, S \geq 0$ and $Q = 2^R 3^S$.*

1. *For every $0 \leq a \leq R$, $0 \leq b \leq S$, and every $t \in I_k$,*

$$P_{a,b}(t) = \left\lfloor \frac{k}{2^{R-a} 3^{S-b}} \right\rfloor. \tag{18}$$

   Consequently the entire carry rectangle is constant on I_k.

2. *Distinct cylinders produce distinct carry rectangles.*

3. *An abstract edge labeling is realized by some $t \in [0,1)$ if and only if it is compatible in the sense of Definition 4.3.*
4. *Hence there are exactly*
$$2^R 3^S = Q \tag{19}$$
compatible $R \times S$ carry rectangles.

Proof. Let $d = 2^{R-a}3^{S-b}$. If $t \in I_k$, then
$$\frac{k}{d} \le 2^a 3^b t < \frac{k+1}{d}.$$
No integer lies strictly between k/d and $(k+1)/d$, and the right endpoint is excluded. Therefore the floor is $\lfloor k/d \rfloor$, proving (18).

At the upper-right vertex, $P_{R,S}(t) = \lfloor Qt \rfloor = k$. Thus the rectangle recovers k, so different cylinders give different rectangles.

Conversely, start with a compatible abstract labeling and set $P_{0,0} = 0$. Define potentials along edges by
$$P_{a+1,b} = 2P_{a,b} + h_{a,b}, \qquad P_{a,b+1} = 3P_{a,b} + v_{a,b}. \tag{20}$$
The plaquette equation says exactly that the two recursions agree around every square. Since every two monotone lattice paths are related by adjacent square interchanges, the potential is path-independent. The digit ranges imply inductively that
$$0 \le P_{a,b} < 2^a 3^b. \tag{21}$$
Put $k = P_{R,S} \in \{0, \dots, Q-1\}$. Running (20) backward is Euclidean division by 2 or 3; it recovers the unique quotients
$$P_{a,b} = \left\lfloor \frac{k}{2^{R-a}3^{S-b}} \right\rfloor.$$
By the first part, the labeling is therefore realized throughout I_k. The bijection with the Q cylinders proves the count. □

Corollary 4.5 (Explicit edge digits)**.** *For the rectangle associated with k, one has*
$$h_{a,b} = \left\lfloor \frac{k}{2^{R-a-1}3^{S-b}} \right\rfloor - 2\left\lfloor \frac{k}{2^{R-a}3^{S-b}} \right\rfloor, \tag{22}$$
$$v_{a,b} = \left\lfloor \frac{k}{2^{R-a}3^{S-b-1}} \right\rfloor - 3\left\lfloor \frac{k}{2^{R-a}3^{S-b}} \right\rfloor. \tag{23}$$
Equivalently, these are the indicated quotient digits modulo 2 and 3.

Proof. Substitute (18) into (14) and (15). □

4.4 Exact entropy collapse

The edge alphabet contains $R(S+1)$ binary horizontal labels and $(R+1)S$ ternary vertical labels. Without compatibility there are
$$2^{R(S+1)}3^{(R+1)S} \tag{24}$$
labelings. Only $2^R 3^S$ survive.

Corollary 4.6 (Exact compatibility fraction and zero area entropy)**.** *The fraction of edge labelings that are compatible is*

$$\frac{2^R 3^S}{2^{R(S+1)} 3^{(R+1)S}} = 6^{-RS}. \tag{25}$$

The logarithm of the number of feasible patterns is

$$R \log 2 + S \log 3, \tag{26}$$

which is boundary-order rather than area-order. In particular the two-dimensional pattern entropy per plaquette tends to zero as $R, S \to \infty$.

This exact collapse is stronger than merely listing the plaquette equations. It shows that the restricted digit equations are independent in aggregate: each of the RS squares removes an exact factor 6 from the naive count.

Example 4.7 (The first nontrivial rectangle)**.** For $R = S = 1$, $Q = 6$. Ordering the four edge labels as $(h_{0,0}, v_{0,0}, h_{0,1}, v_{1,0})$, the six compatible patterns are

k	edge pattern
0	$(0, 0, 0, 0)$
1	$(0, 0, 1, 1)$
2	$(0, 1, 0, 2)$
3	$(1, 1, 1, 0)$
4	$(1, 2, 0, 1)$
5	$(1, 2, 1, 2)$

The plaquette equation is $3h_{0,0} + v_{1,0} = 2v_{0,0} + h_{0,1}$. Of the $2^2 3^2 = 36$ possible edge labelings, exactly six satisfy it.

4.5 Infinite compatible fields

The finite theorem has a useful compact inverse-limit form.

Theorem 4.8 (Global reconstruction from a compatible carry field)**.** *Suppose binary labels $h_{a,b}$ and ternary labels $v_{a,b}$ are given on the entire first quadrant and satisfy every plaquette equation. Reconstruct potentials from $P_{0,0} = 0$ by (20). Then the closed intervals*

$$J_{a,b} = \left[\frac{P_{a,b}}{2^a 3^b}, \frac{P_{a,b} + 1}{2^a 3^b}\right] \tag{27}$$

are compatible under refinement and have a unique common point $t \in [0, 1]$. For $t \in [0, 1)$ the labels are the floor carries of t; the sole additional endpoint field at $t = 1$ is the all-maximal field $h \equiv 1$, $v \equiv 2$.

Proof. The recurrence says that a horizontal child interval is one of the two equal subintervals of its parent, and a vertical child is one of the three equal subintervals. Plaquette compatibility makes these refinements independent of path. Along the diagonal $(a, b) = (n, n)$, the interval length is 6^{-n}, so the nested closed intervals have a unique common point. If $t < 1$, the half-open version of the finite cylinder formula identifies every floor and hence every edge label. If the intersection point is 1, all potentials are $2^a 3^b - 1$, giving the maximal field. □

Status: new finite and inverse-limit core. Lemma 4.1, Proposition 4.2, Theorems 4.4 and 4.8, and Corollary 4.6 are NEW–PROVED. They require only Euclidean division, finite-grid path independence, and nested intervals. They are suitable for direct Lean formalization without any transcendence theory.

5 Furstenberg universality and complete smooth residues

5.1 The external density input

We use the following classical theorem in its special $2, 3$ form.

Theorem 5.1 (Furstenberg; Boshernitzan)**.** *If $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then*

$$\{\{2^r 3^s \alpha\} : r, s \geq 0\} \tag{28}$$

is dense in $[0, 1]$.

Furstenberg proved the result by topological dynamics [3]; Boshernitzan later gave an elementary proof [4]. The theorem is qualitative. Quantitative variants are discussed in Section 8.

A small but important tail observation will be used repeatedly. For any fixed R_0, S_0, the number $2^{R_0} 3^{S_0} \alpha$ is still irrational, so

$$\{\{2^r 3^s \alpha\} : r \geq R_0,\ s \geq S_0\} \tag{29}$$

is also dense. Thus every open interval is visited arbitrarily far out in both coordinates, not merely somewhere in the orbit.

5.2 Every compatible finite carry pattern occurs

Theorem 5.2 (Universal finite carry language)**.** *Fix an irrational α, dimensions R, S, and a compatible $R \times S$ carry rectangle. For every $R_0, S_0 \geq 0$, there exist $r \geq R_0$ and $s \geq S_0$ such that the shifted carry block*

$$\big(h_{r+a,s+b}(\alpha), v_{r+a,s+b}(\alpha)\big)_{0 \leq a \leq R,\ 0 \leq b \leq S} \tag{30}$$

is exactly the prescribed rectangle.

Proof. By Theorem 4.4, the prescribed rectangle corresponds to a unique cylinder I_k of denominator $Q = 2^R 3^S$. Its interior is nonempty. Apply the tail density (29) to choose $r \geq R_0, s \geq S_0$ with $t = \{2^r 3^s \alpha\}$ in that interior. The carries in the shifted block are the carries of t on the $R \times S$ rectangle, hence are the required pattern. □

Corollary 5.3 (Finite-pattern no-go theorem)**.** *The set of finite compatible carry rectangles is independent of the irrational number α. Consequently, no argument of the following form can prove the Alaoglu–Erdős conjecture:*

> *associate to a hypothetical counterexample its $\times 2, \times 3$ carry field, inspect only bounded windows of carry labels, and derive a contradiction by exhibiting a compatible finite pattern that cannot occur.*

Every compatible bounded window already occurs for every irrational exponent.

This does not make the carry field useless. It says precisely what additional data must be retained: vertex heights, rational numerators, reduced denominator exponents, prime support, or quantitative return times. Discarding all of those and keeping only a finite label window loses the arithmetic distinction.

5.3 A complete residue theorem for smooth dilates

The same cylinder bijection immediately yields an arithmetic consequence which seems not to have been isolated in the supplied report.

Theorem 5.4 (Complete floor residues modulo every 3-smooth modulus)**.** *Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and $Q = 2^R 3^S$. Then*

$$\{\lfloor 2^r 3^s \alpha \rfloor \bmod Q : r \geq R,\ s \geq S\} = \mathbb{Z}/Q\mathbb{Z}. \tag{31}$$

Moreover, every residue occurs with r and s arbitrarily large.

Proof. Fix $k \in \{0, \dots, Q-1\}$. By tail density, choose r_0, s_0 arbitrarily large with

$$t = \{2^{r_0} 3^{s_0} \alpha\} \in I_k.$$

Write $2^{r_0} 3^{s_0} \alpha = n + t$. Then

$$\begin{aligned} \left\lfloor 2^{r_0+R} 3^{s_0+S} \alpha \right\rfloor &= \lfloor Q(n+t) \rfloor \\ &= Qn + \lfloor Qt \rfloor = Qn + k. \end{aligned}$$

Thus the floor is congruent to k modulo Q. Since k was arbitrary, all residues occur. Choosing the anchor beyond any prescribed tail proves the final assertion. □

Remark 5.5 (General two-base form). The finite carry classification works verbatim for multiplicatively independent integers $p, q \geq 2$: horizontal digits lie in $\{0, \dots, p-1\}$, vertical digits in $\{0, \dots, q-1\}$, the plaquette equation is

$$q h_{a,b} + v_{a+1,b} = p v_{a,b} + h_{a,b+1},$$

and there are exactly $p^R q^S$ compatible rectangles. Furstenberg density then gives complete residues modulo $p^R q^S$. The bases $2, 3$ are special here only because of the conjecture.

5.4 Why the theorem is stronger than ordinary equidistribution language

The conclusion is not simply that the orbit is dense. A single hit in I_k encodes an entire coherent rectangle of $R(S+1) + (R+1)S$ edge labels and $(R+1)(S+1)$ floor values. Thus one real interval simultaneously prescribes all lower-scale binary and ternary carries in the block. The exact count $2^R 3^S$ explains why: the full rectangle contains no more information than its upper-right floor residue.

This compression will be crucial below. Under a hypothetical counterexample, the same residue k controls two rational systems at once.

6 Arithmetic lift under a hypothetical counterexample

Throughout this section assume the inherited counterexample data (1)–(3).

6.1 The exact block formula

Fix an anchor (r, s) and abbreviate

$$q = 2^r 3^s, \qquad \nu = \lfloor q\beta \rfloor, \qquad t = \{q\beta\}. \tag{32}$$

For local coordinates $a, b \geq 0$, put $m_{a,b} = 2^a 3^b$. Then

$$\nu_{r+a,s+b} = m_{a,b}\nu + P_{a,b}(t), \tag{33}$$

where $P_{a,b}(t) = \lfloor m_{a,b}t \rfloor$ is the carry potential from (13).

Proof. Since $q\beta = \nu + t$,

$$\lfloor m_{a,b}q\beta \rfloor = \lfloor m_{a,b}\nu + m_{a,b}t \rfloor = m_{a,b}\nu + \lfloor m_{a,b}t \rfloor.$$

□

The compact rational pair (5) therefore obeys two synchronized piecewise-monomial recurrences:

$$X_{r+a+1,s+b} = \frac{X^2_{r+a,s+b}}{2^{h_{a,b}(t)}}, \qquad X_{r+a,s+b+1} = \frac{X^3_{r+a,s+b}}{2^{v_{a,b}(t)}}, \tag{34}$$

$$Y_{r+a+1,s+b} = \frac{Y^2_{r+a,s+b}}{3^{h_{a,b}(t)}}, \qquad Y_{r+a,s+b+1} = \frac{Y^3_{r+a,s+b}}{3^{v_{a,b}(t)}}. \tag{35}$$

The same binary and ternary edge digits drive both rational systems. This is the multiplicative two-dimensional analogue of the source report's one-dimensional synchronized Sturmian recurrence.

6.2 Reduced denominator exponents

Recall the structural factorizations

$$M = 2^\lambda W, \quad W \text{ odd}, \qquad N = 3^\mu Z, \quad 3 \nmid Z.$$

Define

$$d^{(2)}_{r,s} = \nu_{r,s} - \lambda q_{r,s}, \qquad d^{(3)}_{r,s} = \nu_{r,s} - \mu q_{r,s}. \tag{36}$$

Then (5) is in lowest terms as

$$X_{r,s} = \frac{W^{q_{r,s}}}{2^{d^{(2)}_{r,s}}}, \qquad Y_{r,s} = \frac{Z^{q_{r,s}}}{3^{d^{(3)}_{r,s}}}. \tag{37}$$

The block formula gives

$$d^{(j)}_{r+a,s+b} = m_{a,b}d^{(j)}_{r,s} + P_{a,b}(t), \qquad j \in \{2, 3\}. \tag{38}$$

Theorem 6.1 (Synchronized complete denominator residues)**.** *Let $Q = 2^R 3^S$. For every residue $k \bmod Q$ and every prescribed tail $r \geq r_0$, $s \geq s_0$, there is an anchor (r, s) beyond that tail such that at the top of the $R \times S$ block,*

$$d^{(2)}_{r+R,s+S} \equiv d^{(3)}_{r+R,s+S} \equiv k \pmod{Q}. \tag{39}$$

Consequently each of the two reduced denominator exponent sequences attains every residue modulo Q, and they do so at the same indices with the same residue.

Proof. Choose an anchor with $t_{r,s} \in I_k$ by Theorem 5.2. At the top vertex, $P_{R,S}(t) = \lfloor Qt \rfloor = k$. Since $m_{R,S} = Q$, (38) gives

$$d^{(j)}_{r+R,s+S} = Q d^{(j)}_{r,s} + k,$$

which proves the congruences. □

Corollary 6.2 (One full-depth branch)**.** *Because* $\min(\lambda, \mu) = 0$*, at least one of the two synchronized denominator exponents is exactly the common floor $\nu_{r,s}$. Thus one branch realizes the complete residue theorem with no cancellation from a structural base factor.*

> **Status: arithmetic significance.** Theorem 6.1 is stronger than applying density separately to the two compact rational coordinates. It asserts synchronized residue universality for two different reduced-denominator systems, with an inherited guarantee that at least one system has maximal depth. Nevertheless, the residue pattern itself is still universal and therefore does not yet contradict integrality.

6.3 The two extreme rectangles

The cylinders I_0 and I_{Q-1} have especially rigid potentials.

Proposition 6.3 (All-zero and all-maximal blocks)**.** *Let $Q = 2^R 3^S$ and $m_{a,b} = 2^a 3^b$.*

1. *If $0 < t < 1/Q$, then*

   $$P_{a,b}(t) = 0, \qquad h_{a,b} = 0, \qquad v_{a,b} = 0 \tag{40}$$

   throughout the rectangle, and

   $$\nu_{r+a,s+b} = m_{a,b}\nu, \quad X_{r+a,s+b} = X_{r,s}^{m_{a,b}}, \quad Y_{r+a,s+b} = Y_{r,s}^{m_{a,b}}. \tag{41}$$

2. *If $1 - 1/Q < t < 1$, then*

   $$P_{a,b}(t) = m_{a,b} - 1, \qquad h_{a,b} = 1, \qquad v_{a,b} = 2 \tag{42}$$

   throughout the rectangle, and

   $$\nu_{r+a,s+b} = m_{a,b}\nu + m_{a,b} - 1, \tag{43}$$

   $$X_{r+a,s+b} = 2\left(\frac{X_{r,s}}{2}\right)^{m_{a,b}}, \qquad Y_{r+a,s+b} = 3\left(\frac{Y_{r,s}}{3}\right)^{m_{a,b}}. \tag{44}$$

Both types of blocks occur arbitrarily far out.

Proof. For $0 < t < 1/Q$ and $m_{a,b} \le Q$, one has $0 < m_{a,b}t < 1$, so all potentials vanish. For $1 - 1/Q < t < 1$,

$$m_{a,b} - 1 < m_{a,b}t < m_{a,b},$$

so $P_{a,b} = m_{a,b} - 1$. The carry labels and rational formulas follow from their definitions. Tail occurrence follows from Furstenberg density. □

The all-zero rectangles will be the quantitative focus. They give exact powers over an entire two-dimensional block, not merely one unusually small fractional part.

7 Synchronized leading-digit gaps and residual arithmetic

The all-zero carry rectangles admit a concrete interpretation in the ordinary base-2 and base-3 expansions of the integral outputs. This section records the interpretation exactly and audits the most tempting attempts to extract a contradiction from it.

7.1 The paired residuals

Fix an anchor (r, s) and abbreviate

$$q = 2^r 3^s, \qquad \nu = \lfloor q\beta \rfloor, \qquad t = \{q\beta\}. \tag{45}$$

Define the positive integral residuals

$$\Delta_2(q) = M^q - 2^\nu, \qquad \Delta_3(q) = N^q - 3^\nu. \tag{46}$$

They satisfy the exact normalized identities

$$\frac{\Delta_2(q)}{2^\nu} = 2^t - 1, \qquad \frac{\Delta_3(q)}{3^\nu} = 3^t - 1. \tag{47}$$

In particular, when $0 < t < 1/Q$,

$$0 < \Delta_2(q) < (2^{1/Q} - 1)2^\nu, \qquad 0 < \Delta_3(q) < (3^{1/Q} - 1)3^\nu. \tag{48}$$

The exponent ν is the same in both inequalities. That synchronization is the counterexample-specific input; a theorem about one base in isolation would not use all the information.

7.2 A simultaneous leading-zero theorem

For $b \in \{2, 3\}$ and $Q \ge 2$, put

$$G_b(Q) = \max\left\{0, \left\lfloor -\log_b(b^{1/Q} - 1)\right\rfloor\right\}. \tag{49}$$

Thus

$$G_b(Q) = \log_b Q - \log_b(\log b) + O_b(Q^{-1}) + O(1), \qquad Q \longrightarrow \infty, \tag{50}$$

where the final $O(1)$ accounts for the integer part.

Theorem 7.1 (Synchronized leading-zero gaps)**.** *Assume the hypothetical counterexample normalization* (1)*. Let* $Q = 2^R 3^S \geq 2$*. There are arbitrarily large* r *and* s *for which, with* q, ν *as in* (45)*, the following hold simultaneously.*

1. *The binary expansion of* M^q *begins with a* 1 *in position* ν*, followed by at least* $G_2(Q)$ *zero digits.*
2. *The ternary expansion of* N^q *begins with a* 1 *in position* ν*, followed by at least* $G_3(Q)$ *zero digits.*
3. *More explicitly,*

$$M^q = 2^\nu + \Delta_2(q), \quad 0 < \Delta_2(q) < 2^{\nu - G_2(Q)}, \tag{51}$$

$$N^q = 3^\nu + \Delta_3(q), \quad 0 < \Delta_3(q) < 3^{\nu - G_3(Q)}. \tag{52}$$

Proof. By Proposition 6.3, choose arbitrarily remote anchors with $0 < t < 1/Q$. The first inequality in (48) and the definition of $G_2(Q)$ give

$$\Delta_2(q) < (2^{1/Q} - 1)2^\nu \leq 2^{\nu - G_2(Q)}.$$

An integer smaller than $2^{\nu - G}$ cannot occupy any binary position from $\nu - 1$ down through $\nu - G$. This proves the binary statement. The ternary argument is identical. For $Q \geq 2$ we also have $3^t < \sqrt{3} < 2$, so the leading ternary digit in position ν is indeed 1, rather than 2. □

Corollary 7.2 (Arbitrarily remote paired gaps)**.** *For every fixed* $G \geq 1$*, a hypothetical counterexample would produce infinitely many smooth indices* $q = 2^r 3^s$ *for which* M^q *has at least* G *binary zero digits immediately after its leading digit and* N^q *has at least* G *ternary zero digits immediately after its leading digit, with the leading positions equal.*

Proof. Take $Q = 2^R 3^S$ large enough that both $G_2(Q)$ and $G_3(Q)$ are at least G, then apply Theorem 7.1. □

> **Status: what is genuinely new here.** Classical digit results can force the number of nonzero digits of powers to grow. That does not forbid a long *leading* gap along a sparse subsequence. The new constraint is paired: the binary and ternary gaps occur at the same smooth power and have the same leading exponent ν. No theorem located in the literature search converts precisely this synchronization into a contradiction.

7.3 Exact tail relation and the denominator tax

Set

$$u_2 = \frac{\Delta_2(q)}{2^\nu}, \qquad u_3 = \frac{\Delta_3(q)}{3^\nu}. \tag{53}$$

Then $1 + u_2 = 2^t$ and $1 + u_3 = 3^t$, so

$$1 + u_3 = (1 + u_2)^\theta, \qquad \theta = \log_2 3, \tag{54}$$

and equivalently

$$\log_{1+u_2}(1 + u_3) = \theta. \tag{55}$$

Both u_2 and u_3 are rational. For an all-zero block, $u_2, u_3 = O(Q^{-1})$. Taylor's theorem therefore gives

$$\left| \frac{u_3}{u_2} - \theta \right| \ll Q^{-1}, \tag{56}$$

with an absolute implied constant once $Q \geq 2$.

At first sight (56) looks like a strong rational approximation to θ. Written in integers, however,

$$\frac{u_3}{u_2} = \frac{2^\nu \Delta_3(q)}{3^\nu \Delta_2(q)}. \tag{57}$$

The numerator and denominator have height exponential in q, whereas the error is only $O(Q^{-1})$. Standard irrationality measures for $\log_2 3$ therefore give no contradiction unless q is extraordinarily small as a function of Q. This is the same denominator tax that reappears in the hitting-time formulation below.

7.4 Divisibility along an all-zero rectangle

The exact scaling in (41) implies a family of exact integral factorizations. If $0 < t < 1/Q$ and $1 \leq m \leq Q$, then $\lfloor mq\beta \rfloor = m\nu$ and

$$\Delta_2(mq) = M^{mq} - 2^{m\nu} = \Delta_2(q) \sum_{j=0}^{m-1} M^{q(m-1-j)} 2^{\nu j}, \tag{58}$$

$$\Delta_3(mq) = N^{mq} - 3^{m\nu} = \Delta_3(q) \sum_{j=0}^{m-1} N^{q(m-1-j)} 3^{\nu j}. \tag{59}$$

Consequently

$$\Delta_2(q) \mid \Delta_2(mq), \qquad \Delta_3(q) \mid \Delta_3(mq) \qquad (1 \leq m \leq Q). \tag{60}$$

One may refine these identities with the homogeneous cyclotomic factors $\Phi_d(A, B)$:

$$\Delta_2(mq) = \prod_{d \mid m} \Phi_d(M^q, 2^\nu), \qquad \Delta_3(mq) = \prod_{d \mid m} \Phi_d(N^q, 3^\nu). \tag{61}$$

These are useful exact data, but they do not themselves improve the smallness exponent. Each quotient in (58) is naturally of size about $m2^{(m-1)\nu}$ or $m3^{(m-1)\nu}$, exactly compensating for the growth of the higher residual. Primitive-divisor theorems add new primes to each branch but do not couple the two branches. A successful use of (60) therefore needs a genuinely joint invariant.

8 Smooth hitting times and the quantitative bottleneck

The preceding conclusions are qualitative because Furstenberg's theorem does not say how large the first multiplier must be. The correct quantitative parameter is as follows.

Definition 8.1 (One-sided smooth hitting time)**.** For irrational α and $Q \geq 2$, define

$$H_\alpha(Q) = \min \left\{ 2^r 3^s : r, s \geq 0, \quad 0 < \{2^r 3^s \alpha\} < \frac{1}{Q} \right\}. \tag{62}$$

Furstenberg density implies that $H_\alpha(Q)$ is finite.

The one-sided definition targets the all-zero rectangle. Replacing it by distance to the nearest integer changes none of the scale comparisons below.

8.1 A polynomial lower bound from logarithmic forms

Theorem 8.2 (Baker lower bound for a hypothetical counterexample)**.** *Assume* (1)*. There exist effectively computable constants* $c_\beta > 0$*,* $\eta_\beta > 0$*, and* $Q_0(\beta)$ *such that*

$$H_\beta(Q) \geq c_\beta Q^{\eta_\beta} \qquad (Q \geq Q_0(\beta)). \tag{63}$$

Proof. Let $q = 2^r 3^s$ occur in the defining set of $H_\beta(Q)$ and put $\nu = \lfloor q\beta \rfloor$. Since M is not a power of 2, the logarithms $\log M$ and $\log 2$ are linearly independent over $\mathbb{Q}$. The nonzero linear form

$$\Lambda = q \log M - \nu \log 2 = \{q\beta\} \log 2 \tag{64}$$

therefore satisfies a standard effective lower bound

$$|\Lambda| \geq c_1 \max\{q, \nu, 3\}^{-\kappa} \geq c_2 q^{-\kappa} \tag{65}$$

for constants $c_1, c_2 > 0$ and $\kappa > 0$ depending only on M; see, for example, Baker–Wüstholz [7]. On the other hand, $0 < \Lambda < (\log 2)/Q$. Rearranging gives $q \geq (c_2 Q/\log 2)^{1/\kappa}$, which has the stated form. □

The exponent in (63) is not expected to be optimal. Its importance is qualitative: it excludes recurrence faster than every fixed power.

8.2 The best available effective upper scale

The number $\beta = \log_2 M$ is Diophantine-generic by the same logarithmic-form estimate. It therefore lies in the scope of effective versions of Furstenberg's theorem. Bourgain–Lindenstrauss–Michel–Venkatesh proved an effective density statement for Diophantine-generic points [5]. Gayfulin and Moshchevitin later gave an elementary explicit formulation whose density radius, in height language, is essentially

$$\big(\log\log\log X\big)^{-1/8+\varepsilon} \tag{66}$$

for smooth multipliers at most a fixed power of X [6].

Proposition 8.3 (Published effective upper scale)**.** *Assume* (1)*. For every* $\varepsilon > 0$ *there is an effective constant* $C_{\beta,\varepsilon} > 0$ *such that, for all sufficiently large* Q*,*

$$H_\beta(Q) \leq \exp\Big(\exp\Big(\exp\Big(C_{\beta,\varepsilon} Q^{8+\varepsilon}\Big)\Big)\Big)\,. \tag{67}$$

Proof. Apply the uniform effective density theorem of Gayfulin–Moshchevitin to a point in the middle of $(0, 1/Q)$, say $1/(2Q)$. A density radius below $1/(2Q)$ guarantees a smooth orbit point in that interval. Solving

$$(\log\log\log X)^{-1/8+\varepsilon'} < \frac{1}{2Q}$$

for X, and slightly enlarging the exponent to absorb ε', yields the triple-exponential expression. Their theorem permits the smooth multiplier to be bounded by a fixed power of X, which is absorbed by changing $C_{\beta,\varepsilon}$. □

Combining the two estimates gives the present quantitative corridor

$$c_\beta Q^{\eta_\beta} \;\le\; H_\beta(Q) \;\le\; \exp\exp\exp\big(C_{\beta,\varepsilon} Q^{8+\varepsilon}\big). \tag{68}$$

The gap between these scales is the principal obstruction exposed by the new route. The exact constants are secondary; any subpolynomial upper bound would already be far more than enough.

8.3 A clean sufficient closing lemma

Research target 8.4 (Subpolynomial smooth recurrence target)**.** Prove that if $M = 2^\beta$ and $N = 3^\beta$ are both integers, then along an unbounded sequence of 3-smooth moduli Q,

$$H_\beta(Q) = Q^{o(1)}. \tag{69}$$

Theorem 8.5 (The target would prove Alaoglu–Erdős)**.** *If Research target 8.4 holds, then every real x with $2^x, 3^x \in \mathbb{Z}$ is an integer.*

Proof. Assume a nonintegral solution and pass to the primitive β. The target gives, for every $\delta > 0$ and infinitely many Q, a smooth $q < Q^\delta$ with $0 < \{q\beta\} < 1/Q$. But Theorem 8.2 gives $q \ge c_\beta Q^{\eta_\beta}$. Taking $\delta < \eta_\beta$ and Q sufficiently large is impossible. □

> **Status: the exact new wall.** Topological dynamics supplies every finite rectangle but no useful first-occurrence bound. Transcendence theory supplies a polynomial exclusion zone but no recurrence. The missing theorem must use simultaneous integrality to improve the recurrence scale. Improving the constants in either existing estimate without crossing the polynomial/subpolynomial boundary will not settle the conjecture by this route.

9 Metric rarity and the topological–arithmetic boundary

The target (69) is not a generic dynamical property. It would place a hypothetical counterexample in an exceptionally recurrent class which is invisible both to finite carry languages and to ordinary metric theory.

Definition 9.1 (Smooth approximation exponent)**.** For irrational α, define

$$\omega_{2,3}(\alpha) = \limsup_{r+s\to\infty} \frac{-\log \|2^r 3^s \alpha\|_{\mathbb{R}/\mathbb{Z}}}{r\log 2 + s\log 3}, \tag{70}$$

where $\|y\|_{\mathbb{R}/\mathbb{Z}}$ is distance to the nearest integer.

Proposition 9.2 (Baker finiteness)**.** *If $\alpha = \log_B A$ for multiplicatively independent positive integers $A, B > 1$, then*

$$\omega_{2,3}(\alpha) < \infty. \tag{71}$$

In particular this holds for the hypothetical β.

Proof. For $q = 2^r 3^s$, a nearest integer p gives the nonzero logarithmic form $q\log A - p\log B$. A polynomial lower bound in q implies $\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \gg q^{-\kappa}$ for a fixed κ, and hence the ratio in (70) is bounded above by $\kappa + o(1)$. □

Theorem 9.3 (Metric value)**.** *For Lebesgue-almost every $\alpha \in [0,1]$,*

$$\omega_{2,3}(\alpha) = 0. \tag{72}$$

Proof. Fix $\sigma > 0$ and let

$$E_{r,s}(\sigma) = \left\{\alpha \in [0,1] : \|2^r 3^s \alpha\|_{\mathbb{R}/\mathbb{Z}} < (2^r 3^s)^{-\sigma}\right\}.$$

Multiplication by an integer preserves Lebesgue measure on the circle, so

$$|E_{r,s}(\sigma)| \le 2(2^r 3^s)^{-\sigma}.$$

The double sum factors and converges:

$$\sum_{r,s\ge 0} |E_{r,s}(\sigma)| \le 2\left(\sum_{r\ge 0} 2^{-\sigma r}\right)\left(\sum_{s\ge 0} 3^{-\sigma s}\right) < \infty.$$

The first Borel–Cantelli lemma therefore says that almost every α belongs to only finitely many $E_{r,s}(\sigma)$. Intersecting over positive rational σ gives $\omega_{2,3}(\alpha) \le 0$. The definition gives the reverse inequality. □

Proposition 9.4 (Subpolynomial hitting is infinite-exponent recurrence)**.** *If* (69) *holds along an unbounded sequence, then*

$$\omega_{2,3}(\beta) = +\infty. \tag{73}$$

Proof. Choose $Q_j \to \infty$ and smooth $q_j \le Q_j^{\varepsilon_j}$, with $\varepsilon_j \to 0$, such that $0 < \{q_j\beta\} < 1/Q_j$. Then

$$\frac{-\log \|q_j\beta\|_{\mathbb{R}/\mathbb{Z}}}{\log q_j} \ge \frac{\log Q_j}{\varepsilon_j \log Q_j} = \varepsilon_j^{-1} \longrightarrow \infty.$$

□

Thus the closing target asks simultaneous integrality to force a recurrence behavior which is not merely measure zero but incompatible with every finite irrationality measure. This explains why a proof cannot come from Furstenberg minimality alone: minimality is shared by every irrational point, whereas the desired recurrence is at the opposite extreme from the metric norm.

9.1 A hierarchy of information

The new approach separates four levels which are easy to conflate.

1. **Finite compatibility:** the plaquette equation and digit ranges. This is completely classified by Theorem 4.4.
2. **Topological recurrence:** every compatible finite pattern occurs. This holds for every irrational point by Theorem 5.2.
3. **Quantitative recurrence:** how soon a specified pattern occurs. This is measured by $H_\alpha(Q)$ and is not controlled by the finite language.

4. **Arithmetic acceleration:** an improvement caused by the simultaneous rational representations $2^\beta = M$ and $3^\beta = N$. This is the missing level.

Any argument which uses only levels 1 and 2 is ruled out by the finite-pattern no-go theorem. Any argument which reaches level 3 but retains the published triple-exponential scale remains far from the Baker boundary.

10 Digit lifts, multiplication carries, and integral defect sequences

A second route compares the ordinary digits of the two integer outputs. It is independent of the two-dimensional floor-carry field, but the two routes meet on the all-zero rectangles.

10.1 The two digit-lift maps

Write a nonnegative integer m in binary and a nonnegative integer n in ternary:

$$m = \sum_{j\ge 0} \varepsilon_j 2^j, \quad \varepsilon_j \in \{0,1\}, \qquad n = \sum_{j\ge 0} d_j 3^j, \quad d_j \in \{0,1,2\}. \tag{74}$$

Define

$$L(m) = \sum_{j\ge 0} \varepsilon_j 3^j, \qquad R(n) = \sum_{j\ge 0} d_j 2^j. \tag{75}$$

Thus L reads a binary digit string in base 3, while R reads a ternary digit string at the value 2. Put

$$\theta = \log_2 3 > 1, \qquad \rho = \log_3 2 = \theta^{-1} \in (0,1). \tag{76}$$

Theorem 10.1 (Sharp direction of the digit lifts)**.** *The following statements hold.*

1. *L is strictly increasing on $\mathbb{N}$.*
2. *For every $m \ge 0$,*
$$L(m) \le m^\theta, \tag{77}$$
with equality exactly when $m = 0$ or m is a power of 2.
3. *More quantitatively, if $m = 2^B + r$ with $0 < r < 2^B$, then*
$$m^\theta - L(m) \ge \left(\frac{3}{2}\right)^B r. \tag{78}$$
4. *For every $n \ge 0$,*
$$n^\rho \le R(n), \tag{79}$$
with equality exactly when $n = 0$ or n is a power of 3.

Proof. For monotonicity, let B be the highest binary position where $m < n$ differ. The contribution 3^B of the new leading digit exceeds the largest possible loss from all lower positions:

$$3^B - \sum_{j=0}^{B-1} 3^j = \frac{3^B + 1}{2} > 0.$$

We prove the quantitative estimate by induction on B. The scalar inequality

$$(1+u)^\theta \geq 1 + u^\theta + u \qquad (0 \leq u \leq 1) \tag{80}$$

holds because the difference has value 0 at both endpoints and has negative second derivative on $(0,1)$; here $2^\theta = 3$ is used at $u = 1$. With $a = 2^B$ and $u = r/a$, induction gives $L(r) \leq r^\theta$, and hence

$$\begin{aligned} m^\theta - L(m) &= (a+r)^\theta - a^\theta - L(r) \\ &\geq (a+r)^\theta - a^\theta - r^\theta \\ &\geq a^{\theta-1} r = \left(\frac{3}{2}\right)^B r. \end{aligned}$$

This is strict for $r > 0$, while repeated equality occurs precisely for powers of 2.

For R, use subadditivity of $x \mapsto x^\rho$:

$$n^\rho = \left(\sum_j d_j 3^j\right)^\rho \leq \sum_j (d_j 3^j)^\rho = \sum_j d_j^\rho 2^j \leq \sum_j d_j 2^j = R(n).$$

Equality requires one nonzero summand and that its digit be 1, which gives a power of 3. □

10.2 Composition and exact valuations

Proposition 10.2 (Composition laws)**.** *For every $m, n \geq 0$,*

$$R(L(m)) = m, \tag{81}$$

while

$$L(R(n)) \geq n. \tag{82}$$

Equality in (82) holds exactly when the ternary normalization of n requires no binary carries after its digits are read at 2; in particular it holds whenever all ternary digits are 0 or 1.

Proof. The ternary expansion of $L(m)$ consists of the binary digits of m, all belonging to $\{0,1\}$, so reading it back at 2 gives m. For the other direction, start with the ternary digit polynomial $\sum d_j X^j$. Its value at 2 is $R(n)$. Normalizing the coefficients in base 2 replaces two units in position j by one unit in position $j+1$. When evaluated at 3, each such carry changes $2 \cdot 3^j$ into 3^{j+1} and therefore increases the value by 3^j. The initial value at 3 is n and the normalized value is $L(R(n))$. □

Proposition 10.3 (Valuation transfer)**.** *For every positive integer m,*

$$v_3(L(m)) = v_2(m). \tag{83}$$

For every positive integer n,

$$v_2(R(n)) \geq v_3(n), \tag{84}$$

with equality if the least nonzero ternary digit of n is 1.

Proof. If the least binary 1 of m occurs in position k, then $L(m) = 3^k(1+3u)$ for an integer u. If the least nonzero ternary digit of n is d_k, then $R(n) = 2^k(d_k + 2u)$; the parenthesis is odd when $d_k = 1$ and is even when $d_k = 2$. □

10.3 Multiplication carries

The maps are not ring homomorphisms, but their failure has a definite sign and an exact carry expansion.

Theorem 10.4 (Signed multiplicativity and carry gains)**.** *For all nonnegative integers a, b,*

$$L(ab) \geq L(a)L(b), \tag{85}$$

$$R(ab) \leq R(a)R(b). \tag{86}$$

More precisely, there are finitely supported nonnegative integer sequences $\kappa_j(a, b)$ and $\lambda_j(a, b)$ such that

$$L(ab) - L(a)L(b) = \sum_{j \geq 0} \kappa_j(a, b)3^j, \tag{87}$$

$$R(a)R(b) - R(ab) = \sum_{j \geq 0} \lambda_j(a, b)2^j. \tag{88}$$

The coefficients count elementary carries in a normalization algorithm.

Proof. Multiply the binary digit polynomials of a and b. Before normalization, their value at 3 is $L(a)L(b)$. Every elementary base-2 carry replaces two units at position j by one at position $j + 1$ and increases the value at 3 by exactly 3^j. After all carries the polynomial is the binary expansion of ab, whose value at 3 is $L(ab)$. Summing the carry gains proves (87). The ternary case is analogous: replacing three units at position j by one at $j + 1$ decreases evaluation at 2 by exactly 2^j. □

Remark 10.5. The strictness conditions are concrete. Equality in (85) means that the convolution of the two binary digit strings needs no carry; equality in (86) means the corresponding ternary convolution needs no carry. This packages multiplication of output powers into another nonnegative integer cocycle.

10.4 Consequences for a hypothetical output pair

For $n \geq 1$, define

$$D_n = N^n - L(M^n), \qquad E_n = R(N^n) - M^n. \tag{89}$$

Theorem 10.6 (Positive integral digit defects)**.** *Under (1), D_n, E_n are positive integers for every $n \geq 1$. Moreover,*

$$\frac{N^n}{3} \leq L(M^n) < N^n, \qquad M^n < R(N^n) < 4M^n, \tag{90}$$

and hence

$$\lim_{n\to\infty} L(M^n)^{1/n} = N, \qquad \lim_{n\to\infty} R(N^n)^{1/n} = M. \tag{91}$$

Proof. Because M is not a power of 2, neither is M^n, so Theorem 10.1 gives $L(M^n) < M^{n\theta} = N^n$. The reverse inequality is strict because N^n is not a power of 3. If $B = \lfloor n\beta \rfloor$, then the leading binary digit of M^n is in position B, so $L(M^n) \geq 3^B = N^n/3^{\{n\beta\}} \geq N^n/3$. The ternary expansion of $N^n < 3^{B+1}$ uses positions at most B, whence $R(N^n) \leq 2\sum_{j=0}^{B} 2^j < 2^{B+2} \leq 4M^n$. Taking nth roots proves the limits. □

Thus the digit-lift route cannot win by a fixed exponential gap: the two lifted sequences have exactly the conjectural exponential rates. The useful information is in their subexponential defects.

Let

$$B_n = \lfloor n\beta \rfloor, \quad t_n = \{n\beta\}, \quad r_n = M^n - 2^{B_n}, \quad s_n = N^n - 3^{B_n}. \tag{92}$$

Then $0 < r_n < 2^{B_n}$ and $0 < s_n < 2 \cdot 3^{B_n}$.

Theorem 10.7 (Tail inequalities and a sharp defect sandwich)**.** *For every $n \geq 1$,*

$$L(r_n) < s_n, \tag{93}$$

and

$$3^{B_n}(2^{t_n} - 1) \leq D_n \leq 3^{B_n}(3^{t_n} - 1). \tag{94}$$

If $t_n < \rho = \log_3 2$, then the leading ternary digit of N^n is 1 and

$$r_n < R(s_n). \tag{95}$$

In that range,

$$(1-\rho)2^{B_n}(3^{t_n} - 1)^{\rho} \leq E_n < 4\,2^{B_n}(3^{t_n} - 1)^{\rho}. \tag{96}$$

Proof. Since the leading binary digit of M^n is isolated, $L(M^n) = 3^{B_n} + L(r_n)$. Subtracting from $N^n = 3^{B_n} + s_n$ gives $D_n = s_n - L(r_n) > 0$, proving (93). The quantitative lower bound (78), with $r_n = 2^{B_n}(2^{t_n} - 1)$, gives the lower half of (94). The upper half follows from $L(M^n) \geq 3^{B_n}$.

If $t_n < \rho$, then $N^n < 2 \cdot 3^{B_n}$, so its leading ternary digit is 1 and $R(N^n) = 2^{B_n} + R(s_n)$. Hence $E_n = R(s_n) - r_n > 0$. Put $u = 3^{t_n} - 1 \in (0,1)$. By (79), concavity of x^{ρ}, and $u \leq u^{\rho}$,

$$\begin{aligned} E_n &\geq s_n^{\rho} - 2^{B_n}\left((1+u)^{\rho} - 1\right) \\ &\geq 2^{B_n}(u^{\rho} - \rho u) \geq (1-\rho)2^{B_n}u^{\rho}. \end{aligned}$$

For the upper bound, $E_n < R(s_n)$. If the highest ternary position of s_n is J, then $R(s_n) \leq 2\sum_{j=0}^{J} 2^j < 4 \cdot 2^J \leq 4s_n^{\rho}$, which gives (96). □

For an all-zero smooth anchor $n = q$ with $t_q < 1/Q$, the theorem gives the paired relative estimates

$$\frac{D_q}{N^q} = \Theta(t_q) = O(Q^{-1}), \qquad \frac{E_q}{M^q} = \Theta(t_q^{\rho}) = O(Q^{-\rho}), \tag{97}$$

with effective constants independent of the anchor. The differing exponents reflect the Hölder distortion between ternary and binary digit evaluation.

Proposition 10.8 (Defect valuation transfer)**.** *Let $\lambda = v_2(M)$ and $\mu = v_3(N)$. If $\lambda \neq \mu$, then for every $n \geq 1$,*

$$v_3(D_n) = n\min\{\lambda, \mu\}. \tag{98}$$

In particular, under the primitive condition (3), if exactly one of λ, μ is positive, then every D_n is a 3-adic unit.

Proof. By Proposition 10.3, $v_3(L(M^n)) = v_2(M^n) = n\lambda$, while $v_3(N^n) = n\mu$. When these valuations differ, the valuation of their difference is the smaller one. □

10.5 An exact bridge from digit carries to analytic defects

Define the nonnegative carry functionals

$$C_2(a,b) = L(ab) - L(a)L(b), \qquad C_3(a,b) = R(a)R(b) - R(ab). \tag{99}$$

Theorem 10.9 (Defect–carry recurrences)**.** *For all positive integers m, n,*

$$C_2(M^m, M^n) = N^m D_n + N^n D_m - D_m D_n - D_{m+n}, \tag{100}$$

$$C_3(N^m, N^n) = M^m E_n + M^n E_m + E_m E_n - E_{m+n}. \tag{101}$$

Proof. Write $L(M^k) = N^k - D_k$ in $C_2(M^m, M^n) = L(M^{m+n}) - L(M^m)L(M^n)$ and expand. The second identity follows by writing $R(N^k) = M^k + E_k$. □

Corollary 10.10 (Carry sparsity inside an all-zero block)**.** *Suppose $0 < t_q < 1/Q$ and $m + n \le Q$. Then*

$$0 \le C_2(M^{mq}, M^{nq}) \ll \frac{m+n}{Q} N^{(m+n)q}, \tag{102}$$

$$0 \le C_3(N^{mq}, N^{nq}) \ll \left(\frac{m}{Q}\right)^{\rho} M^{(m+n)q} + \left(\frac{n}{Q}\right)^{\rho} M^{(m+n)q}, \tag{103}$$

with absolute implied constants.

Proof. For $k \le Q$, one has $t_{kq} = kt_q < 1$. Apply Theorem 10.7 to bound $D_{kq}/N^{kq} \ll kt_q \ll k/Q$ and, when $kt_q < \rho$, $E_{kq}/M^{kq} \ll (kt_q)^{\rho}$. If one of $m, n, m+n$ lies outside that smaller range, enlarge the absolute constant using (90). Substitution in Theorem 10.9 gives the result. □

The recurrence is a concrete point of contact between the two new packages. A hypothetical counterexample forces multiplication of enormous powers to incur only a small high-base carry gain along the same smooth subsequences that create the floor carry rectangles. At present, general lower bounds for these multiplication-carry functionals are too weak: the first unavoidable carry may occur far below the leading position, exactly as permitted by the leading-zero gaps.

11 Stress tests: what the new information does not prove

A useful progress report should not merely accumulate true statements; it should identify which combinations have actually been tested and why they stop. The table below records the principal audits.

Attempt	Exact input	Failure point or surviving target
Forbid a finite carry pattern	Plaquette compatibility and the counterexample carry field	Every compatible finite pattern occurs for every irrational exponent. This strategy is ruled out by Corollary 5.3.

Attempt	Exact input	Failure point or surviving target
Use the rarity of an all-zero rectangle	Arbitrarily long simultaneous leading gaps from Theorem 7.1	Such rectangles also occur for every irrational orbit. The missing information is the first hitting scale, not existence.
Approximate $\log_2 3$ by the residual ratio	Equation (56)	The rational denominator is exponential in the smooth multiplier q. An $O(Q^{-1})$ error is far too weak unless $q = Q^{o(1)}$.
Exploit ordinary cyclotomic factors	Equations (58)–(61)	The quotient factors have exactly the size needed to absorb the residual growth. Primitive primes occur separately in the two branches and are not synchronized.
Use a theorem on sparse digits of powers	Paired leading gaps	Known single-base sparsity results force many nonzero digits globally, not the absence of an arbitrarily long leading gap on a sparse subsequence. A joint theorem with the same leading position would be needed.
Use supermultiplicativity of L	$L(M^{m+n}) \geq L(M^m)L(M^n)$	The exponential growth rate is exactly N by (91); there is no fixed exponential deficit to amplify.
Use the carry gains C_2, C_3	The exact recurrences (100)–(101)	The first mandatory multiplication carry can lie far below the leading digit. Current lower bounds match the scale allowed by the leading gaps.
Use complete residues modulo smooth Q	Theorem 6.1	Residue universality is itself a consequence of density and is compatible with every irrational point. One needs a height, valuation, or factorization consequence attached to a selected residue.
Replace density by equidistribution	The orbit $\{2^r 3^s \beta\}$	Furstenberg's theorem gives density, not pointwise equidistribution for every irrational. Assuming normality or uniform discrepancy would insert an unproved theorem.

Attempt	Exact input	Failure point or surviving target
Apply the effective $\times 2, \times 3$ theorem directly	Proposition 8.3	The available upper scale is roughly triple exponential in a power of Q, while Baker excludes only a polynomial scale.
Infer a theorem from the reported rival breakthrough	Current announcements	No public Alaoglu–Erdős statement, proof object, or Lean source was found. The claim cannot be used as a lemma.

11.1 Three false shortcuts in formula form

Density is not a rate. The implication

$$\overline{\{2^r 3^s \beta : r, s \geq 0\}} = \mathbb{T} \quad \Longrightarrow \quad H_\beta(Q) \leq Q^C$$

is false as a matter of logic. Compactness proves finiteness for each fixed Q but supplies no uniform dependence on Q.

A small real residual need not have a large structural valuation. When M is odd, $\Delta_2(q) = M^q - 2^\nu$ is odd, even when it is tiny relative to 2^ν. Thus archimedean smallness does not automatically create 2-adic depth. The analogous statement holds for the ternary branch.

Two small forms are not independent forms. The expressions

$$q \log M - \nu \log 2 = t \log 2, \qquad q \log N - \nu \log 3 = t \log 3$$

are proportional because of the original logarithmic relation. Multiplying their individual Baker lower bounds does not create a two-dimensional determinant; doing so would disguise the four-exponentials barrier rather than cross it.

12 Computational reconnaissance

The finite classification is exact and does not require computation. Computation is still useful for checking indexing conventions, exploring first-occurrence scales, and preparing formal tests. All calculations in this section are labeled COMPUTED and are logically dispensable.

12.1 Exact enumeration of compatible rectangles

For a fixed R, S, the following integer-only pseudocode constructs every compatible rectangle. It is a direct executable version of Theorem 4.4.

Listing 1: Exact enumeration from the cylinder index

```
def carry_rectangle(R: int, S: int, k: int):
```

```
    Q = (2**R) * (3**S)
    assert 0 <= k < Q

    P = {}
    for a in range(R + 1):
        for b in range(S + 1):
            denominator = (2 ** (R - a)) * (3 ** (S - b))
            P[a, b] = k // denominator

    h = {(a, b): P[a + 1, b] - 2 * P[a, b]
         for a in range(R) for b in range(S + 1)}
    v = {(a, b): P[a, b + 1] - 3 * P[a, b]
         for a in range(R + 1) for b in range(S)}

    assert all(x in (0, 1) for x in h.values())
    assert all(x in (0, 1, 2) for x in v.values())
    assert all(3*h[a, b] + v[a + 1, b]
               == 2*v[a, b] + h[a, b + 1]
               for a in range(R) for b in range(S))
    return P, h, v
```

Running over $0 \le k < 2^R 3^S$ produces exactly that many distinct pairs (h, v). No floating-point operation occurs.

12.2 First complete cylinder coverage in bounded boxes

A separate high-precision scan asked when a bounded orbit box had hit every cylinder I_k for $Q = 6, 36, 216$. The exponents were restricted to $0 \le r, s \le 60$ and 120 decimal digits were used. The quantity reported is the largest, over all cylinders, of the logarithm of the first multiplier $2^r 3^s$ found for that cylinder.

α	$Q = 6$	$Q = 36$	$Q = 216$
$\log_2 3$	4.852030264	21.48756260	40.80500268
$\log_2 5$	3.871201011	15.77248612	42.76666119
$\sqrt{2}$	4.852030264	19.70935416	52.07880764
π	2.484906650	9.416378455	40.56943661

All cylinders were found within the scan box in these twelve cases. The data show large fluctuations and no stable power law at these tiny scales. In particular, they do not support extrapolation to the target (69). Their purpose is only to validate the cylinder indexing and to illustrate that first complete coverage can be controlled by a very different orbit point for nearby examples.

12.3 Reproducible diagnostic script

Listing 2: High-precision bounded orbit scan

```
import mpmath as mp

mp.mp.dps = 120
alphas = {
```

```
    "log_2(3)": mp.log(3) / mp.log(2),
    "log_2(5)": mp.log(5) / mp.log(2),
    "sqrt(2)": mp.sqrt(2),
    "pi": mp.pi,
}

for R, S in [(1, 1), (2, 2), (3, 3)]:
    Q = (2**R) * (3**S)
    print("Q", Q)
    for name, alpha in alphas.items():
        first = [None] * Q
        for r in range(61):
            for s in range(61):
                q = (2**r) * (3**s)
                t = mp.frac(mp.mpf(q) * alpha)
                k = int(mp.floor(Q * t))
                candidate = (mp.log(q), r + s, r, s)
                if first[k] is None or candidate[0] < first[k][0]:
                    first[k] = candidate
        assert all(item is not None for item in first)
        worst = max(first, key=lambda item: item[0])
        print(name, mp.nstr(worst[0], 12),
              max(item[1] for item in first))
```

A proof-oriented implementation should replace transcendental floating-point comparisons by interval enclosures and record the distance from every sampled point to the nearest cylinder boundary. None of the theorem proofs above depend on this scan.

13 Lean formalization blueprint

The finite carry theory is unusually suitable for formalization because its core can be expressed entirely with natural-number division and remainders. The analytic and external dynamical inputs can be attached later.

13.1 Proposed module graph

Module	Principal contents
`Carry.Basic`	Fractional-part maps, one-step carries, and the cocycle identity.
`Carry.Rectangle`	Potentials, edge labels, plaquette compatibility, and path independence.
`Carry.Cylinders`	Integer quotient formula and the exact Q-pattern bijection.
`Carry.Global`	Nested intervals and inverse-limit reconstruction.
`Carry.Furstenberg`	Interface theorem stating tail density of the $\times 2, \times 3$ orbit.
`Counterexample.Smooth`	Formulas for $\nu_{r,s}$, rational coordinates, reduced denominator exponents, and complete residues.
`Counterexample.Gaps`	Residual bounds, digit gaps, and hitting times.
`Digits.Lift`	Definitions of L, R, power inequalities, composition, and valuations.
`Digits.MulCarry`	Signed multiplicativity and the carry-gain expansions.
`Counterexample.Defects`	D_n, E_n, sandwiches, valuations, and carry recurrences.
`Measure.SmoothBC`	Almost-everywhere vanishing of $\omega_{2,3}$.

The dependency graph is

$$\texttt{Basic} \to \texttt{Rectangle} \to \texttt{Cylinders} \to \begin{cases} \texttt{Global}, \\ \texttt{SmoothLift} \leftarrow \texttt{Furstenberg}, \end{cases}$$

with the digit modules forming a second branch that meets `LeadingGaps` in `Defects`.

13.2 Finite theorem signatures

The following signatures are schematic Lean 4, intended to fix interfaces rather than claim compilation in the present environment.

Listing 3: Schematic finite carry API

```
def potential (R S a b k : Nat) : Nat :=
  k / (2^(R-a) * 3^(S-b))

def hCarry (R S a b k : Nat) : Nat :=
  potential R S (a+1) b k - 2 * potential R S a b k

def vCarry (R S a b k : Nat) : Nat :=
  potential R S a (b+1) k - 3 * potential R S a b k

structure CarryRectangle (R S : Nat) where
  h : Fin R -> Fin (S+1) -> Fin 2
  v : Fin (R+1) -> Fin S -> Fin 3
  plaquette : forall a b,
    3 * (h a b : Nat) + (v a.succ b : Nat) =
    2 * (v a b : Nat) + (h a b.succ : Nat)

theorem cylinder_equiv_carryRectangle (R S : Nat) :
  Fin (2^R * 3^S) <-> CarryRectangle R S
```

A convenient proof strategy is to define the forward map by quotient digits and the inverse map by the upper-right potential. Path independence can be proved either by induction on rectangles or by showing that adjacent path swaps preserve the result. The quotient formula then avoids most real-floor reasoning.

13.3 Schematic digit API

Listing 4: Schematic digit-lift statements

```
def lift23 (m : Nat) : Nat :=
  (Nat.digits 2 m).eval 3

def lift32 (n : Nat) : Nat :=
  (Nat.digits 3 n).eval 2

theorem lift23_le_rpow (m : Nat) :
  (lift23 m : Real) <= (m : Real) ^ (Real.log 3 / Real.log 2)

theorem lift32_ge_rpow (n : Nat) :
  (n : Real) ^ (Real.log 2 / Real.log 3) <= (lift32 n : Real)

theorem lift23_mul_super (a b : Nat) :
  lift23 a * lift23 b <= lift23 (a*b)

theorem lift32_mul_sub (a b : Nat) :
  lift32 (a*b) <= lift32 a * lift32 b

theorem lift32_lift23 (m : Nat) :
  lift32 (lift23 m) = m
```

The formal proof should separate the combinatorial carry-normalization lemma from the real-power inequalities. That makes the signed multiplicativity theorem reusable for general pairs of bases.

13.4 External interfaces and present library boundary

As of 22 August 2026, public mathlib documentation contains substantial APIs for floors, fractional parts, measure theory, and Borel–Cantelli, and it contains density results for additive irrational rotations on the circle [13, 14, 15]. The search used for this report did not locate a public formalization of Furstenberg's multiplicative $\times 2, \times 3$ density theorem. Nor is the full Baker–Wüstholz theorem currently a routine imported dependency; a public Lean evaluation problem still presents it with a placeholder proof [16].

Accordingly, the recommended formalization discipline is:

1. formalize the finite carry bijection and digit-lift package with no axioms;
2. state Furstenberg density behind a single clearly marked interface theorem;
3. state the needed polynomial logarithmic-form bound behind a second interface;
4. derive every counterexample consequence from those two interfaces;
5. replace the interfaces only when independently reviewed formalizations become available.

This yields a useful machine-checked finite core without pretending that the deep external theorems have already been imported.

13.5 Suggested pull-request order

1. `Carry/Cylinders.lean`: quotient digits, plaquettes, exact cardinality.
2. `Digits/Lift.lean`: order, power bounds, compositions, valuations.
3. `Digits/MulCarry.lean`: normalization carries and signed products.
4. `Carry/Realize.lean`: floor realization on half-open intervals.
5. `Counterexample/FiniteConsequences.lean`: block and residue formulas, parameterized by an abstract density hypothesis.
6. `Measure/SmoothBC.lean`: the metric theorem, using mathlib's Borel–Cantelli infrastructure.

The first three items are finite or elementary and can be reviewed independently of the open conjecture.

14 Prioritized research program

The new analysis narrows the next work to a small number of sharply testable statements. They are ranked by potential mathematical leverage rather than ease.

14.1 Priority 1: denominator-sensitive acceleration

The decisive target remains Research target 8.4. A more structured version is the following.

Research target 14.1 (Paired rational-orbit acceleration)**.** Let $M, N \geq 2$ be integers and suppose $\log_2 M = \log_3 N = \beta \notin \mathbb{Q}$. Prove that the rational orbit points

$$\left(\frac{M^q}{2^{\lfloor q\beta \rfloor}}, \frac{N^q}{3^{\lfloor q\beta \rfloor}}\right), \qquad q = 2^r 3^s,$$

enter the box $[1, 2^{1/Q}) \times [1, 3^{1/Q})$ with $q = Q^{o(1)}$ along an unbounded smooth sequence Q.

This formulation exposes the exact arithmetic unavailable for a generic point: both coordinates are rational, their numerator exponents are the same smooth q, and their denominator exponents are equal.

14.2 Priority 2: a joint leading-gap theorem

Research target 14.2 (Matched leading-gap rigidity)**.** Rule out, for multiplicatively independent output pairs $(M, 2)$ and $(N, 3)$ linked by the same β, simultaneous representations

$$M^q = 2^\nu + \Delta_2, \qquad N^q = 3^\nu + \Delta_3$$

with $q = 2^r 3^s$, the same ν, and

$$\Delta_2 < 2^\nu/Q, \qquad \Delta_3 < 3^\nu/Q$$

for a recurrence scale q below a fixed power of Q.

Single-base digit theorems are insufficient. The proof would have to use the common leading position, perhaps through a mixed resultant, a common prime divisor of suitably normalized residuals, or a subspace theorem with a genuinely coupled height.

14.3 Priority 3: high-position multiplication carries

The carry recurrences suggest a finite combinatorial target. Let $\deg_3 C_2(a, b)$ denote the highest position carrying positive weight in (87), and define the analogous quantity for C_3.

Research target 14.3 (Paired high-carry lower bound)**.** Use the fact that $a = M^q$ and $b = N^q$ are synchronized powers to prove that at least one of the multiplication normalizations for $(M^q)^m$ or $(N^q)^m$ incurs a carry within $o(\log Q)$ positions of the leading digit, uniformly for some bounded $m \geq 2$, whenever q begins an all-zero rectangle of modulus Q.

A carry that high would give a lower bound conflicting with Corollary 10.10. The unpaired statement is false because a number may start with an arbitrarily long zero gap.

14.4 Priority 4: exploit selected smooth denominator residues

Theorem 6.1 lets one choose the common top correction $k \bmod Q$ arbitrarily. The next useful theorem would attach a height or valuation cost to special choices, for example $k = 0$, 1, or $Q - 1$. Candidate invariants are:

- the Kummer class of the reduced rational point modulo Qth powers;
- the Smith profile of the numerator–denominator exponent pair;
- primitive prime divisors of the paired residuals;
- the least nonzero multiplication-carry position of the digit lifts.

A residue theorem without one of these costs is already exhausted by density.

14.5 Priority 5: sharpen effective dynamics only where it crosses Baker

A general improvement from triple exponential to double exponential would be substantial dynamics, but it would not close the present argument. The strategically relevant milestone is a polynomial upper bound $H_\beta(Q) \ll Q^C$ with an exponent C that can be compared to the explicit Baker lower exponent. The decisive milestone is subpolynomial. Computational and analytic effort should report results in these terms rather than only in terms of a smaller density-radius constant.

14.6 Priority 6: formalize the finite core now

The carry bijection and digit-lift package are self-contained, nontrivial, and useful regardless of the conjecture. Formalizing them now would prevent indexing errors, make the no-go theorem reusable, and give a stable target against which any claimed breakthrough can be tested. In particular, a rival Lean development should be asked which theorem bridges finite compatibility to arithmetic acceleration; the module graph in Section 13 makes that question precise.

15 General two-base carry rectangles

The finite combinatorics is not special to 2 and 3. Recording the general form clarifies which part of the approach is universal and which part uses the arithmetic of the conjecture.

Let $a, b \geq 2$ be integers. For $t \in [0, 1)$ define

$$P_{i,j}(t) = \left\lfloor a^i b^j t \right\rfloor,$$

with horizontal digits

$$h_{i,j} = P_{i+1,j} - aP_{i,j} \in \{0, \dots, a-1\}$$

and vertical digits

$$v_{i,j} = P_{i,j+1} - bP_{i,j} \in \{0, \dots, b-1\}.$$

Theorem 15.1 (General rectangular carry theorem)**.** *For an $R \times S$ rectangle the edge digits satisfy*

$$bh_{i,j} + v_{i+1,j} = av_{i,j} + h_{i,j+1}. \tag{104}$$

Conversely, every edge labeling in the indicated digit ranges satisfying all plaquette equations is realized on a unique cylinder

$$\left[\frac{k}{a^R b^S}, \frac{k+1}{a^R b^S}\right), \qquad 0 \leq k < a^R b^S.$$

Consequently there are exactly $a^R b^S$ compatible rectangles, and their fraction among all edge labelings is

$$(ab)^{-RS}. \tag{105}$$

Proof. The two paths around a plaquette give

$$P_{i+1,j+1} = abP_{i,j} + bh_{i,j} + v_{i+1,j}$$

after moving horizontally first, and

$$P_{i+1,j+1} = abP_{i,j} + av_{i,j} + h_{i,j+1}$$

after moving vertically first. This proves (104). The converse repeats the path-independence proof of Theorem 4.4. On the cylinder indexed by k,

$$P_{i,j} = \left\lfloor \frac{k}{a^{R-i} b^{S-j}} \right\rfloor.$$

The compatible count is $a^R b^S$, whereas the unconstrained count is $a^{R(S+1)} b^{(R+1)S}$; their quotient is $(ab)^{-RS}$. □

For general multiplicatively independent a, b, Furstenberg's theorem again makes the finite compatible language universal. What is special in Alaoglu–Erdős is that the same exponent produces integral outputs at the two original bases and hence the paired residual and digit-lift structures of Sections 7 and 10.

16 Conclusion

This continuation develops a new multiplicative-orbit framework without repeating the determinant, interpolation, p-adic, and additive-word machinery of the supplied report. The exact progress is:

1. a complete classification of every finite $\times 2, \times 3$ floor-carry rectangle and its exact entropy collapse;
2. a Furstenberg-based theorem that every compatible finite pattern, and every smooth floor residue, occurs for every irrational exponent;
3. the resulting no-go theorem for all bounded-window carry strategies;
4. synchronized complete residue classes for the two reduced denominator exponents of a hypothetical counterexample;
5. arbitrarily long paired leading-zero gaps in powers of the two integer outputs;
6. a precise polynomial-to-triple-exponential corridor for the first such block;
7. a clean subpolynomial hitting-time lemma which would close the conjecture;
8. a digit-lift theory with strict power inequalities, signed multiplicativity, valuation transfer, positive defect sequences, and exact defect–carry recurrences;
9. a finite Lean architecture separating elementary combinatorics from the two deep external inputs.

The strongest negative conclusion is as important as the positive ones: finite symbolic compatibility is completely universal. The proof cannot be hidden in a local carry pattern, because every irrational exponent has the same finite carry language. The strongest positive conclusion is the location of the missing bridge: simultaneous integrality must force an arithmetic acceleration of a topologically generic orbit, or an equivalent high-position carry/leading-gap contradiction.

No full proof is claimed. The report leaves one central quantitative target rather than a diffuse collection of analogies. Any purported breakthrough can now be stress-tested by asking where it proves one of the following genuinely new facts: subpolynomial smooth recurrence, matched leading-gap rigidity, or a paired high-position carry lower bound. Without such a bridge, the argument remains on the universal side of the Furstenberg boundary.

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